

	(1)	(2)	(3)	(4)	(5)
Lower bound	−0.195 (0.064)	−6.665 (1.366)			
Upper bound	0.283 (0.114)	2.771 (2.067)			
Point estimate			−1.025 (1.182)	−0.069 (1.349)	0.954 (1.588)
units	Log(Hours)	Hours	Hours	Hours	Hours
c			0	0.25	0.5

Table 5: Bounds and point estimates for the intensive margin treatment effect in Carranza et al. (2022)

Note: This table shows bounds and point estimates of the intensive margin treatment effect in Carranza et al. (2022), i.e. the treatment effect on hours worked for “always-takers” who would work positive hours regardless of treatment status. The first two columns of the table show Lee (2009) bounds for the effect of treatment on the always-takers when the outcome is log(Hours) and weekly hours, respectively. Columns 3 through 5 show point estimates for the effect on weekly hours worked for always-takers under the assumption that average hours worked by “compliers” (who work only when treated) are 100% lower than for the always-takers. Standard errors are calculated via a non-parametric bootstrap using 1,000 draws, clustered at the assessment date level.

Original specification and sensitivity to units. Sequeira (2016) has repeated cross-sectional data with information on the bribe amount Y_{it} paid on shipment i in year t . She estimates the regression specification

$$\log(1 + Y_{it}) = \beta_0 + D_i \times \text{Post}_t \beta_1 + D_i \beta_2 + \text{Post}_t \beta_3 + X'_{it} \beta_4 + \epsilon_{it}, \quad (6)$$

where D_i is an indicator for whether shipment i is for a product type affected by the tariff change in 2008, Post_t is an indicator for whether year t is after the tariff change, and X_{it} is a vector of covariates related to shipment i in period t . Sequeira (2016) estimates (6) with Y_{it} measured in 2007 Mozambican Metical (MZN) and obtains $\hat{\beta}_{1,(\text{MZN})} = -3.7$ (SE = 1.1). However, estimating the same specification with Y_{it} measured in thousands of U.S. dollars instead yields an estimate of $\hat{\beta}_{1,(\$1000)} = -0.11$ (SE = 0.070).³⁷ These results reinforce the conclusion from Section 2 that treatment effects for $m(y) = \log(1 + y)$ should not be interpreted as approximating a percentage effect.

In what follows, we discuss a variety of alternative approaches that may be reasonable in this context. We note that in a non-experimental setting like this, different approaches may rely on different identifying assumptions. We therefore explicitly discuss the identifying assumptions needed by each of the methods we discuss.

³⁷We use the conversion rate of 1 USD = 24.48 MZN as of January 1, 2007, as provided by fxtop.com.

Proportional treatment effects. One natural approach here is to target the average proportional treatment effect on the treated,

$$\theta_{\text{ATT}\%} = \frac{E[Y_{it}(1) \mid D_i = 1, \text{Post}_t = 1] - E[Y_{it}(0) \mid D_i = 1, \text{Post}_t = 1]}{E[Y_{it}(0) \mid D_i = 1, \text{Post}_t = 1]}.$$

This is the percentage change in the average outcome for the treated group in the post-treatment period.

Identification of $\theta_{\text{ATT}\%}$ requires us to infer the counterfactual post-treatment mean outcome for the treated group, $E[Y_{it}(0) \mid D_i = 1, \text{Post}_t = 1]$. Of course, one approach to obtain such identification would be to assume parallel trends in levels. However, given that the treated and control groups have different pre-treatment means (see the bottom panel of Table 6), it may be unreasonable to expect that time-varying factors (e.g. the macroeconomy) have equal level effects on the outcome. An alternative identifying assumption is to impose that, in the absence of treatment, the *percentage* changes in the mean would have been the same for the treated and control group. As in Wooldridge (2023), this can be formalized using a “ratio” version of the parallel trends assumption,

$$\frac{E[Y_{it}(0) \mid D_i = 1, \text{Post}_t = 1]}{E[Y_{it}(0) \mid D_i = 1, \text{Post}_t = 0]} = \frac{E[Y_{it}(0) \mid D_i = 0, \text{Post}_t = 1]}{E[Y_{it}(0) \mid D_i = 0, \text{Post}_t = 0]}. \quad (7)$$

Intuitively, (7) states that if the treatment had not occurred, the average percentage change in the mean outcome for the treated group would have been the same as the average percentage change in the mean outcome for the control group. Under (7), we can thus estimate the counterfactual percentage change in the mean outcome for the treated group using the observed percentage change for the control group.

Table 6 shows that the sample mean of the outcome for the treated group decreased by 75% between the pre-treatment and post-treatment periods (from 4,742 to 1,172 (MZN)). Under the ratio parallel trends assumption (7), this suggests that the mean outcome for the treated group would also have decreased by 75% in the absence of treatment, thus implying an estimate of 2,602 for the counterfactual mean outcome for the treated group. The actual post-treatment mean for the treated group is 465, which is 82% below this implied counterfactual. This implies that the tariff reduction reduced the average bribe in the post-treatment period by 82%, i.e. $\hat{\theta}_{\text{ATT}\%} = -0.82$. Conveniently, this estimate can also be obtained using Poisson QMLE to estimate

$$Y_{it} = \exp(\beta_0 + D_i \times \text{Post}_t \beta_1 + D_i \beta_2 + \text{Post}_t \beta_3) \epsilon_{it} \quad (8)$$

and then computing $\hat{\theta}_{\text{ATT}\%} = \exp(\hat{\beta}_1) - 1 = -0.82$, as shown in column (1) of Table 6.

We can also re-incorporate the covariates X_{it} by estimating

$$Y_{it} = \exp(\beta_0 + D_i \times \text{Post}_t \beta_1 + D_i \beta_2 + \text{Post}_t \beta_3 + \beta_4' X_{it}) \epsilon_{it}, \quad (9)$$

which yields an estimate of $\theta_{\text{ATT}\%}$ of -0.72 , as shown in the second column of Table 6. As formalized

	(1)	(2)
Post $\times$ Treatment	−1.722 (0.632)	−1.272 (0.606)
Prop. Effect	−0.821 (0.113)	−0.720 (0.170)
Covariates	N	Y
Treated Group Means (Pre, Post):	10527	465
Treated Group Means (Pre, Post):	4742	1172

Table 6: Poisson regression estimates of $\theta_{\text{ATT}\%}$

Note: this table shows Poisson regression estimates of (8) and (9) in columns (1) and (2), respectively. The first row of the table shows the estimate $\hat{\beta}_1$. The second row shows $\exp(\hat{\beta}_1) - 1$, which is the implied estimate of the proportional treatment effect $\theta_{\text{ATT}\%}$. The coefficients on control variables are omitted for brevity. Standard errors are clustered at the four-digit product code as in the original paper. The mean bribe amounts (in MZN) by treatment group and time period are displayed in the bottom panel. The pre-period refers to the year 2007, whereas the post-treatment period is an average over the years 2008, 2011, and 2012 (the three post-treatment years for which data is available).

in Wooldridge (2023), this estimate will be a consistent estimate of $\theta_{\text{ATT}\%}$ if (7) holds conditional on X_{it} , and the conditional expectation of Y_{it} takes the functional form implied by (9) (assuming ϵ_{it} has mean 1 conditional on the covariates). The approach with covariates thus suggests that the tariff change reduced the average bribe for treated products by 72% in the post-treatment period.

Sequeira (2016)’s data only contains information on one year prior to treatment (2007), and so in this context it is not possible to evaluate the plausibility of (7) using periods prior to the policy change of interest. If multiple pre-treatment periods were available, however, one could estimate a Poisson QMLE event-study of the form

$$Y_{it} = \exp \left(\lambda_t + D_i \beta_2 + \sum_{r \neq -1} D_i \times [\text{RelativeTime}_t = r] \beta_r^{ES} \right) \epsilon_{it}, \quad (10)$$

where $\text{RelativeTime}_t = t - 2008$ is the time relative to the treatment date. The event-study coefficients β_r^{ES} for $r < 0$ are analogous to “pre-trends” coefficients in typical difference-in-differences event-studies, and are informative about whether the pre-treatment analogue to (7) holds.³⁸

Log effects with calibrated extensive margin value. The analysis above presented estimates of $\theta_{\text{ATT}\%}$, the proportional change in the *average* bribe caused by the treatment. It is well-known that averages can be heavily influenced by observations in the tail, especially when the outcome has

³⁸More precisely, the exponentiated coefficients $\exp(\hat{\beta}_r) - 1$ correspond to the implied “placebo” proportional treatment effects for periods before treatment. We recommend plotting the exponentiated coefficients in event-studies, although we note that $\exp(\beta) - 1 \approx \beta$ for $\beta \approx 0$. As with typical tests for pre-trends, one should be cautious that a failure to reject the null that the pre-treatment coefficients equal zero does not necessarily imply that the identifying assumption is satisfied (Kahn-Lang and Lang, 2020; Roth, 2022). One can (partially) address these issues by applying sensitivity analysis tools for event-studies (e.g. Rambachan and Roth, 2023) to estimates of (10) to further gauge the robustness of the findings to violations of the identifying assumptions. We also refer the reader to Wooldridge (2023) for extensions of the Poisson regression approach to settings with staggered treatment timing.

a skewed distribution, as is the case here (see [Figure 2](#)). One might argue that a world in which most products receive medium-sized bribes is more corrupt than one in which a very small fraction of products receive large bribes—even if they both produce the same average bribe amount. This motivates studying the treatment effect on a concave transformation of the outcome that is less heavily influenced by outcomes in the tail of the distribution. As an illustration of this, we first normalize the outcome so that 1 corresponds to the value of the minimum non-zero bribe in the data (that is, we divide by $y_{\min} = \min_{Y_{it} > 0} Y_{it} = 15.68$ MZN). We then estimate the treatment effect for the transformed outcome $m(Y)$, where $m(y) = \log(y)$ for $y > 0$ and $m(0) = -x$ for some choice of x , as described in [Section 4.2](#). If x is set to 0, then this estimates the treatment effect in logs where all zero bribes are set to equal the smallest positive bribe in the data; this specification thus “shuts off” the extensive margin change between 0 and $y_{\min}$. If instead x is set to 0.1, for example, then a change between 0 and $y_{\min}$ is valued as the equivalent of a 10 log point change along the intensive margin.

We estimate the treatment effect for these transformations using the analogue to (6) that replaces $\log(1 + Y_{it})$ with $m(Y_{it})$ on the left-hand side.³⁹ The results for $x \in \{0, 0.1, 1, 3\}$ are shown in [Table 7](#). Column (1) shows an effect of 249 log points ($\hat{\beta}_1 = -2.49$) when we treat zero bribes as if they were equal to $y_{\min}$ (i.e. setting $x = 0$). The estimated treatment effect grows in magnitude as we place more value on the extensive margin by increasing x . Interestingly, the original estimate in [Sequeira \(2016\)](#) of -3.748 using $\log(1 + Y)$ is similar to what we obtain when we value a change from 0 to $y_{\min}$ at 300 log points ($x = 3$). The original specification can thus be viewed as placing a rather large weight on the extensive margin.

	(1)	(2)	(3)	(4)
Post $\times$ Treatment	-2.493 (0.740)	-2.538 (0.752)	-2.949 (0.861)	-3.860 (1.106)
Extensive margin value (x):	0.000	0.100	1.000	3.000

Table 7: Explicit calibration of the extensive margin in [Sequeira \(2016\)](#)

Note: this table shows estimates of the treatment effect on the treated using $m(Y)$ as the outcome in [Sequeira \(2016\)](#), where $m(y)$ is defined to equal $\log(y)$ for $y > 0$ and $-x$ for $y = 0$. The outcome is normalized so that $Y = 1$ corresponds to the minimum non-zero value of the outcome. Thus, the treatment effect assigns a value of $100x$ log points to an extensive margin change between 0 and the minimum non-zero value of Y . The treatment effects are estimated using (6), except replacing $\log(1 + Y_{it})$ with $m(Y_{it})$. Standard errors are clustered at the four-digit product code as in the original paper.

5.3 An IV setting: [Berkouwer and Dean \(2022\)](#)

[Berkouwer and Dean \(2022\)](#) conduct an RCT in Nairobi in which they randomize the price for

³⁹As usual, identification of the treatment effect for $m(Y)$ using difference-in-differences requires parallel trends for $m(Y(0))$. The identifying assumption thus varies depending on the choice of x . The results in [Roth and Sant’Anna \(2023\)](#) imply that parallel trends will hold for all values of x when a parallel trends assumption is satisfied for the distribution of $Y(0)$. If more pre-treatment periods were available, these identifying assumptions could be partially evaluated using pre-trends tests. See [Remark 9](#) for additional discussion of identification.